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BE Capstone Project · Data-Driven State Estimation with Embedded Feasibility Validation · VESIT, Mumbai · Automation and Robotics · 2027

BATTERY STATE ESTIMATOR: AN ESN–BASED ALTERNATIVE TO EKF AND COULOMB COUNTING

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Introduction, Objectives & Solution

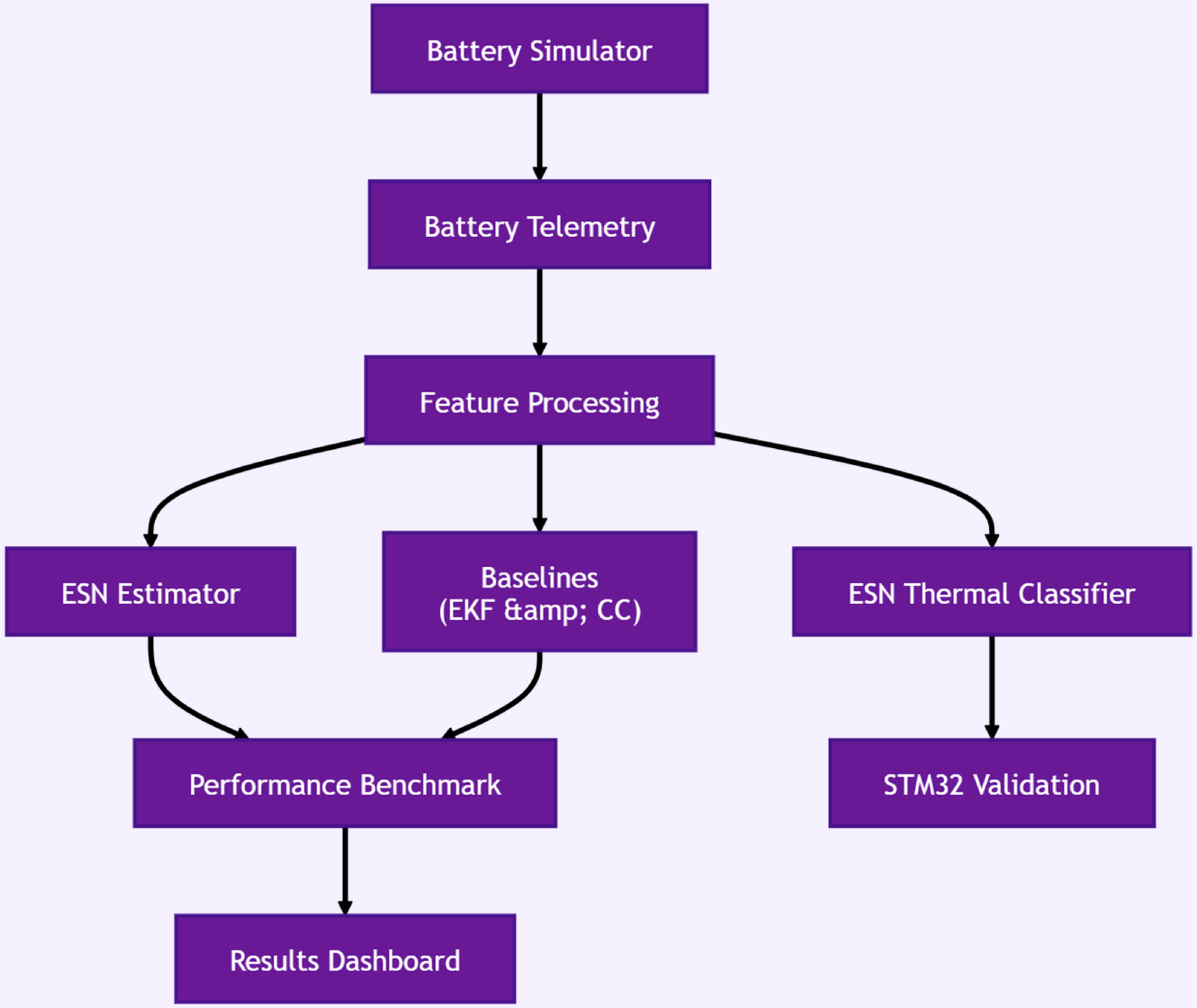
01 **Problem Statement:** Existing BMS estimation relies on Coulomb Counting and EKF, both accumulate drift, depend on precise parameterization and OCV-SOC curves and degrade as the cell ages. This project's core contribution is an Echo State Network (ESN) estimator designed as a direct, lighter-weight replacement for these classical observers and not just a supplement to them.

02 **Core Objectives:** Develop an ESN-based SOC/SOH estimator capable of matching or exceeding EKF and Coulomb Counting accuracy under dynamic EV drive-cycle loads. Use EKF and Coulomb Counting strictly as baseline benchmarks to quantify the ESN's improvement and not as part of the final deployed pipeline. Prove the ESN's computational footprint is small enough to eventually replace these observers on constrained hardware

03 **Solution:** The proposed solution trains an ESN estimator directly on simulated battery telemetry, using EKF and Coulomb Counting purely as accuracy baselines rather than deployment components. A separate ESN classifier variant is compiled to C99 and validated on STM32 hardware to confirm the reservoir-computing approach is deployable under real embedded constraints, while the SOC/SOH estimator itself remains software-validated, with hardware porting reserved for future work.

04 **Validation Scope:** Hardware deployment (STM32) exists solely to test whether the ESN algorithm is deployable in a real-time, low-power setting, it is a feasibility check, not the project's end product. Success is defined by algorithmic accuracy and efficiency, not by the hardware demo itself

System Architecture



Flow Description

- Battery Simulator generates realistic battery telemetry (voltage, current and temperature) to provide consistent input data for model development and evaluation.
- Feature Processing prepares the telemetry and feeds it to the proposed ESN Estimator, benchmark algorithms (EKF & Coulomb Counting) and the ESN Thermal Classifier.
- Performance Benchmarking compares the ESN estimator against the baseline methods to evaluate estimation accuracy, robustness and computational efficiency.
- Validation & Visualization deploys the ESN thermal classifier on the STM32 for embedded feasibility testing, while the Results Dashboard presents comparative performance metrics and outcomes.

Hardware Layer & Edge Diagnostics

STM32 Nucleo used only to simulate and time-profile the ESN under embedded constraints, confirms deployability, doesn't define project scope

2-RC ECM circuit model exists purely to produce physically realistic simulated telemetry for non hardware training of the ESN, it is a data-generation tool

CSR sparse storage and Q12/Q15 fixed-point arithmetic are algorithm-efficiency techniques applied to the ESN, demonstrating it can be compressed enough to replace EKF/CC on real MCUs

This confirms the ESN architecture is deployable on constrained hardware, validated here for the thermal classifier, porting the SOC/SOH estimator to STM32 remains future work

Software Implementation

Primary Deliverable: ESN Estimator (Reservoir Computing)

- SOC: Echo State Network using a fixed random reservoir with fading-memory projection, the proposed production estimator, not a supplementary model
- SOH: Trained SOH-ESN, proposed reservoir-computing replacement for resistance-growth RLS tracking
- Reservoir computing is the core algorithmic choice: only the readout layer is trained, keeping the ESN cheap enough to eventually replace EKF/Coulomb Counting on embedded hardware

Baseline Methods (benchmarking only)

- SOC baseline: Coulomb Counting, Adaptive EKF
- SOH baseline: Online RLS resistance-growth tracking
- Implemented purely to generate a fair, side-by-side accuracy comparison against the ESN

Supporting Infrastructure

- Physics Simulator: Flask service generating 2-RC ECM telemetry that both the ESN and baselines are tested against
- Visualiser Dashboard: Flask application presenting ESN vs. EKF vs. Coulomb Counting comparison charts as the primary analytical output
- Database Layer: MongoDB Atlas (with in-memory fallback) storing telemetry and estimator results, infrastructure to run the comparison, not part of the algorithmic contribution

Testing, Results & Outlook

Key Result: The ESN is benchmarked against EKF and Coulomb Counting baselines, targeting sub-1.5% SOC RMSE and sub-1.0% SOH RMSE. Edge metrics, measured on the ESN classifier variant, demonstrate the reservoir-computing approach is deployable on embedded hardware, supporting the case for eventually porting the SOC/SOH estimator as well. 51 automated tests validate the estimator and classifier pipelines.

Advantages: ESN avoids EKF's dependency on precise parameter models and Coulomb Counting's unbounded drift, while staying lighter than LSTM/GRU alternatives, making real-time replacement viable on low-power MCUs. It also generalizes across drive cycles without re-deriving electrochemical parameters for each condition and its fixed reservoir with a trained readout keeps retraining cost low when adapting to new cell chemistries.

Future Scope: Port the trained SOC/SOH ESN estimator (currently exported as esn_estimator_weights.h) into STM32 firmware, following the same CSR/fixed-point path already validated for the classifier. Fully replace EKF/CC with ESN in production (currently retained only as benchmarks) and add online adaptive readout tuning, real hardware-in-the-loop validation and extend the approach to multi-cell packs for broader BMS applicability.

Conclusion

- The project's deliverable is a data-driven ESN estimator positioned to replace EKF and Coulomb Counting, not a hardware product or a three-way hybrid system
- EKF and Coulomb Counting were implemented only to serve as rigorous baselines for benchmarking the ESN's accuracy, rather than as part of the deployed pipeline
- STM32 hardware testing validated that the ESN's reservoir-computing architecture (proven via the thermal classifier) is deployable within real BMS constraints, porting the SOC/SOH estimator itself to firmware remains future work
- This positions ESN-based estimation as a viable, lighter-weight alternative to classical observers in next-generation BMS design

References

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